

Full-length article

Optimal energy management system for a community microgrid: A case study from the Gambia

Maimuna Drammeh^{a,*}, Ravi Samikannu^{b,c}, Abid Yahya^b, Maruliya Begam Kadarmydeen^d, Kavitha R^e, Kanimozhi Gunasekaran^f, Meeradevi T^g, Gladys Gamariel^b

^a Pan African University of Water and Energy Sciences (PAUWES), Tlemcen, Algeria

^b Botswana International University of Science and Technology, Palapye, Botswana

^c Department of Electronics and Communication Engineering, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, India

^d Periyar Maniammai Institute of Science & Technology, Thanjavur, India

^e Department of Mathematics and Statistics, Faculty of Science and Humanities, SRM Institute of Science and Technology, Kattankulathur Campus, Chengalpattu, India

^f Centre for Smart Grid Technologies, Vellore Institute of Technology, Chennai, India

^g Department of ECE, Kongu Engineering College, Erode, Tamil Nadu, India

ARTICLE INFO

Handling Editor: Janice A. Beecher

Keywords:

Microgrids
Rural electrification
Solar photovoltaic
Wind turbine
Generator
Battery storage

ABSTRACT

Remote communities often lack access to reliable electricity. This study investigates the feasibility of a microgrid system tailored for Kantong Kunda, a rural community in The Gambia. The community's current energy consumption and demand are determined through data collection using the Epicollect5 survey tool to characterize the local energy consumption and demand profile accurately. HOMER Pro software was employed to simulate and optimize hybrid microgrid configuration, prioritizing both cost-effectiveness and environmental sustainability. The proposed system integrates 79.8 kW of Solar Photovoltaic (SPV), a 60-kW diesel generator, 374 batteries, and a 22.8 kW converter. The optimized design yields a Net Present Cost (NPC) of \$251,474.80 and a Levelized Cost of Energy (LCOE) of \$0.08527/kWh, which is well below the region's grid electricity tariff. This configuration yields 16.1 % excess electricity, a 10.5 % Return on Investment (ROI), a 14.2 % Internal Rate of Return (IRR), and a 6.03-year payback period, while cutting total emissions by 133,981.4 kg compared to a diesel-only baseline. The work contributes a microgrid design designed for rural African communities, and the findings demonstrate that microgrids can deliver reliable, affordable, and low-carbon electricity through decentralized energy systems for remote communities.

1. Introduction

Approximately 80 % of those without access to electricity worldwide live in Sub-Saharan Africa (SSA) (International Energy Agency, 2022). The Gambia had 63.70 % access to electricity by 2021, with 69 % access in urban areas, while rural areas had only 28 % (International Trade Administration). Historically, the country has relied on heavy fuel oil imports for electricity generation (MOPE, 2021). Extending the national grid to small remote communities with minimal demand would be costly and inefficient, requiring substantial time and financial resources. Transmitting electricity over long distances from generation centres would pose economic and environmental challenges, including increased transmission losses and higher greenhouse gas emissions from burning more fossil fuels to meet overall demand. Therefore,

decentralized renewable energy generation to meet local demand offers significant financial and environmental benefits, including reduced emissions (Qureshi et al., 2024). One such decentralized system is microgrids.

Microgrids, formed by low or medium-voltage intelligent distribution networks comprising diverse distributed generation sources like wind generators, PV generation, thermal power plants, and micro-turbines, and energy storage devices, alongside controllable loads, can accelerate electricity access to rural communities or areas where the central electricity grid cannot reach in the short or medium term. They offer an alternative generation solution for providing adequate electricity (Liu et al., 2017), (Hartono et al., 2013).

However, research on their feasibility in The Gambia's specific context, particularly within individual communities, remains limited. This study addresses this gap by focusing on the Kantong Kunda

* Corresponding author.

E-mail address: maimunadrammeh10@gmail.com (M. Drammeh).

<https://doi.org/10.1016/j.jup.2025.102132>

Received 8 May 2025; Received in revised form 9 December 2025; Accepted 10 December 2025

Available online 24 December 2025

0957-1787/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature			
AC	Alternating Current	HOMER	Hybrid Optimization Model for Electric Renewables
BES	Battery Energy Storage	HRES	Hybrid Renewable Energy System
BESS	Battery Energy Storage System	HRS	Hydrogen Refuelling Station
CC	Capital Cost	IPPs	Independent Power Producers
CO ₂	Carbon dioxide	IRR	Internal Rate of Return
COE	Cost of Electricity	kW	Kilowatt
COH	Cost of Hydrogen	kWh	Kilowatt-hour
CSP	Concentrated Solar Plant	LCOE	Levelized Cost of Energy
DC	Direct Current	m/s	Meters per Second
DERs	Distributed Energy Resources	MPPT	Maximum Power Point Tracking
DG	Diesel Generator	NASA	National Aeronautics and Space Administration
DG or DGs	Distributed Generation or Distributed Generations	NPC	Net Present Cost
ESS	Energy Storage Systems	O&M	Operation and Maintenance
EMS	Energy Management System	PV	Photo Voltaic
GDP	Gross Domestic Product	RE	Renewable Energy
HESS	Hybrid Energy Storage System	ROI	Return on Investment
HES	Hybrid Energy System	SSA	Sub-Saharan Africa
		Yr	Year

community.

This study also fills a research gap in renewable electricity accessibility in typical rural areas of The Gambia by examining the technical and economic criteria for designing the most cost-effective and reliable hybrid energy system. The findings will serve as a guide and framework for future hybrid energy system implementation, helping to promote environmental and socioeconomic development through affordable, sustainable electricity.

2. Literature review

Access to electricity is crucial for the social and economic development of nations (Zhang et al., 2019), especially those that are still developing. The global increase in energy prices impacts countries like The Gambia that depend on energy imports. The Gambia struggles to generate and transmit energy to meet the demand, with a projected increase of nearly 140 MW in new capacity by 2025 (MOPE, 2021).

Renewable energy presents a significant opportunity for the Gambia. It has an estimated solar resource potential of 428 MW of solar PV and 953 MW of CSP (IRENA, 2018). Lahmeyer International (2006), Singh et al. (2013) and Blodgett and Marett (2015), suggests that CSP may not be suitable for the country due to its relatively small power requirement. While (Marong et al., 2018) suggests that a CSP thermal plant could meet the demand from industry and households. The average solar radiation across all its regions ranges from 4.4 to 6.7 kWh/m²/day. The estimated resource potential for wind power, biomass, and small hydropower projects is 44 MW, 60 MW, and 12 MW, respectively. Large-scale hydro projects are not feasible (IRENA, 2018). In inland areas, wind speeds are typically moderate, measuring less than 4.0 m/s². However, due to the westward movement of wind from the sea, it can rise to approximately 4.3 m/s² at a height of 30 m (Lahmeyer International, 2006)– (Blodgett and Marett, 2015). (Marong et al., 2018) emphasizes that, in addition to existing oil-based systems, the Gambia needs to develop its renewable energy capacity and hydroelectric imports to achieve a more secure, cost-effective, and environmentally sustainable energy supply.

The integration of renewable energy sources requires distributed generation technology, which enables efficient use of all energy sources (Tariq et al., 2024). Microgrids provide the flexibility to select precise renewable power sources and quantities to be integrated into the system while improving energy reliability, security, and resilience against high-impact rare events (Sabzpoosh Saravi et al., 2024), (Chandak and Rout, 2021). It generates and controls power, manages the system's load

demand, and ensures system stability. The energy storage unit within a microgrid serves as a supplementary energy source, helping stabilize variable energy sources and dynamic loads. It facilitates load-sharing operations and mitigates surges and electrical disturbances (Hartono et al., 2013). Due to their widespread availability, lead-acid and lithium-ion batteries are increasingly used to manage microgrids (Jirdehi et al., 2020). Operational efficiency and reliability are enhanced using MG control and management systems. The energy management system manages power generation, forecasts load demand, compares it to actual demand, controls reactive power flow, and reduces energy consumption. It facilitates optimal operation by establishing seamless communication and control among various components, including the energy management controller (e.g., SCADA), power supply systems, PV, Wind, and other renewable energy sources, and non-conventional energy supply systems such as generators and switching systems (Tran et al., 2021; Zia et al., 2018; Kowalczyk et al., 2015). Many loads can participate in energy management through load response or demand response. Specific loads can shift their consumption from peak to off-peak times as defined in an agreement between the user and the network provider. An Energy Management Strategy (EMS) is required, as having multiple energy sources creates significant disadvantages and new challenges for effectively regulating energy flow in the supply system. When used in standalone systems, the EMS guarantees load supply continuity under all circumstances, optimizes the use of renewable energy sources, integrates with the optimization problem to reduce the Cost Of Energy (COE) production, guards against overloading-related component damage, and boosts power system stability. In grid-connected systems, the energy management approach leverages low-cost tariff periods and manages energy flow to and from the grid to shift peak loads and support metering. A central controller is crucial for implementing this strategy and may also integrate with a monitoring system, such as a SCADA system. (Olatomiwa et al., 2016).

HOMER (Hybrid Optimization of Multiple Energy Resources) is an optimization tool that evaluates system size and energy management by comparing economic and technical options and accounting for input data uncertainties. It simulates annual energy system performance hourly, identifying optimal configurations, sizing renewable resources, and ensuring demand satisfaction within constraints. HOMER tests various system designs, performs hourly energy balance calculations, and manages energy flow between components, including decisions on generator operation and battery charging/discharging (Olatomiwa et al., 2016), (Hossein Shahinzadeh et al., 2016). In this study, HOMER Pro's optimization feature serves as the exclusive Energy Management

Strategy for the hybrid microgrid. The program automatically addresses the power dispatch problem using its patented algorithm, which lowers lifecycle costs while accounting for critical constraints, such as battery depth-of-discharge limits, minimum load ratios for generators, and renewable energy curtailment criteria. HOMER's Energy Management System (EMS) prioritizes renewable energy use through time-series simulations, activates diesel generators only during energy deficits, and maintains system stability through integrated load-following and cycle-charging techniques. The platform's sensitivity analysis tools evaluate the system's resilience to variable solar and wind resources and load changes, enabling deterministic optimization without relying on external algorithms. Although HOMER produces optimal component sizing and dispatch schedules, real-world implementation would require integration with physical controllers, such as PLCs or SCADA systems, for real-time execution of these strategies (Lambert et al., 2006). Microgrids serve as an alternative solution for remote communities. Several studies on the technical, economic, and environmental feasibility have been conducted in various regions around the world, though few have been conducted in Africa. Barber et al. (2019), Alshakhs and Arefifar (2020), Veilleux et al. (2020), Xu et al. (2019) and Murty and Kumar (2020) are some studies conducted outside Africa, and the results show the technical, economic, and environmental feasibility of microgrids in those regions.

A study conducted by (Madduri et al., 2016) used field data from Kenya and India to support the design of a scalable microgrid architecture. The study revealed that the proposed microgrid, which utilized Direct Current (DC), had a Levelized Cost of Electricity (LCOE) of less than \$0.40 per kilowatt-hour, and the experimental results demonstrated the overall efficiency of a locally installed prototype of a DC microgrid. In a study by Kuznetsov et al. (2019), the potential to meet the energy needs of Abu-Mongar, a rural community in Egypt, was assessed, and the simulation results demonstrated that the proposed hybrid microgrid system is economically feasible and offers a practical solution for electrification in the area. A case study by Elkadeem et al. (2022) found that a PV, wind turbine, and converter system connected to the grid is the most optimal and economically viable option, based on optimization results for eight energy alternatives in Egypt at the Sokhna Industrial Zone in Suez. It has an NPC and LCOE that are 17 % and 22 % lower than the existing grid, respectively, serving as the base case system. A study by Nnaji et al. (2019) focusing on rural electrification in Nigeria found that using LED bulbs for their proposed smart microgrid significantly reduces NPC by 56 %–81 % compared to a system that solely relies on a diesel generator and incandescent light bulbs.

Microgrids have been extensively studied for electricity generation in rural areas and implemented in various remote regions worldwide. However, to the authors' knowledge, there is little research in West Africa regarding the technical, economic, and environmental feasibility of microgrids for rural electrification, particularly in the Gambian context, especially in the Kantong Kunda community. Additionally, little research has discussed the effects of the minimum battery life on the design, economics, and environmental factors of the microgrid.

3. Problem statement and case study area: Kantong Kunda Community

The electricity supply is inadequate, plagued by instability, load shedding, and blackouts. In 2020, the poverty rate in rural areas, where agriculture is predominant, was estimated at 76 % (Bellagamba, 2020). Kantong Kunda is a community primarily engaged in farming. Despite needing electricity for their livelihood, they have no access to the national grid. The community is in the Kiang West District of the Lower River Region in Gambia, at Latitude 13.3342° N and Longitude 15.9575° W. It has a population of approximately 460 people, with the majority being female, living in 29 households. The community has a primary school, a mosque, and small businesses that make up its commercial sector. It lacks many social amenities, such as hospitals or any other

public administration offices, necessitating travel to Kiang Keneba, 6.9 km away, for health care. The nearest community, Kiang Jali, 2.4 km away, is a larger settlement with grid electricity. The drinking water source is solar water boreholes within the community.

About 55 % of households had access to electricity from solar power, while the remaining 45 % relied on battery-powered lamps for night lighting. Among the 16 households with solar home systems, 5 provided only lighting, while the other 11 were used for both lighting and charging mobile phones. Only one household had a functioning refrigerator, which operated only during the day. Another household had a television but lacked sufficient solar energy. Mobile charging and other electrical loads were used during the day, from 10 a.m. to 5 p.m., to ensure households could use their lights throughout the night. Fig. 1 shows the Kantong Kunda Community with outlined locations.

3.1. Energy resources assessment

3.1.1. Solar radiation

Kantong Kunda has an annual solar irradiation average of about 5.45 kWh/m²/day. The clarity of the atmosphere, indicated by a mean clearness index of 0.572 throughout the year, suggests some overcast and moderate attenuation of solar radiation.

3.1.2. Wind speed

The monthly average wind speed of Kantong Kunda at 50m above the earth's surface over 30 years was derived from Homer Pro's in-built NASA POWER database. The wind speed shows monthly variation, with the highest average in January at 5.99 m/s and the lowest in October at 3.21 m/s. The mean annual wind speed is 4.63 m/s.

3.1.3. Temperature

The monthly average temperature of Kantong Kunda over 30 years from the NASA POWER database shows that the highest average temperatures occur from March to June, with May recording the highest at 30.9 °C and January the lowest at 23.34 °C.

3.2. Methodology

This study was carried out using the HOMER Pro software. The site location was obtained using GPS and entered into HOMER Pro. Satellite data on wind speed, solar radiation, and temperature were then obtained to determine resource availability. The energy demand of the case study site was estimated using information gathered from the questionnaire created for the research. The components of the microgrid were selected based on available resource data, local market availability, and pricing from both local and internet sources (for components not available locally). A comprehensive simulation and optimization considering technical, economic, and environmental aspects was performed. HOMER Pro considers the Photo Voltaic (PV) system Maximum Power Point Tracker (MPPT), a device that continuously adjusts the PV array's operating point to deliver maximum power under varying environmental conditions. The software uses a simplified linear model to estimate PV output based on solar radiation, which closely aligns with the performance of systems equipped with MPPT. This assumption enhances the accuracy of the simulation results and reflects standard practice in modern PV system design.

3.3. Mathematical modelling and simulation

3.3.1. Modelling of power generation, storage, and loads

Considering A as the number of PV modules, η_{PV} as the module efficiency, η_{wire} as wiring efficiency G_{STC} and T_{STC} as the reference solar irradiance and temperature of PV cell at Standard conditions (1000W/m² and Temperature of 25 °C respectively), β_T as thermal factor, and $G_T(t)$ represents the ambient solar radiation (W/m²), $T_{ambient}$ as the ambient Temperature and T_{Test} as the temperature observed during the

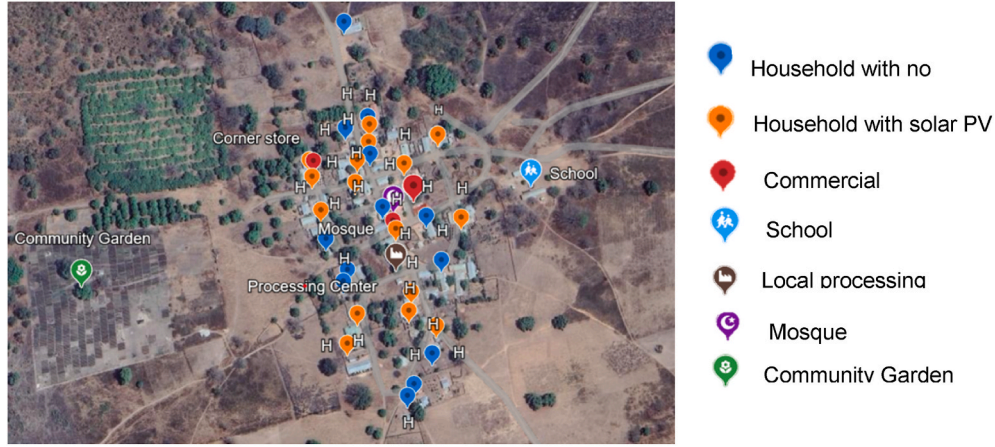


Fig. 1. Map of Kantong Kunda Community with outlined locations.

testing of the modules, then the energy generated from the solar PV module (P_{PV}) and the cell temperature (T_{cell}) can be expressed as (Zaki Diab et al., 2019), (Alilou et al., 2025):

$$P_{PV}(t) = \frac{G_T(t)}{G_{STC}} \times PP_{PV_rated} \times A \times \eta_{PV} \times \eta_{wire} (1 - \beta_T + (T_{cell}(t) - T_{STC})) \quad (1)$$

$$T_{cell}(t) - T_{ambient} = G(t) \times \frac{T_{Test}}{800} \quad (2)$$

3.3.2. Modelling of the wind power system

Given the intermittent nature of wind and variability of wind speeds throughout the day, the expected power output from a wind turbine can be calculated with respect to the measured wind speeds at a desired hub height as follows, with consideration to the cut-in and cut-out speed of the wind turbine:

$$P_W(t) = \begin{cases} [(P_R / (V_r - V_{ci}^3)) \times V^3(t)] - b \times P_R, & V_{ci} < V < V_r \\ P_R, & V_r < V < V_{co} \\ 0, & V < V_{ci} \text{ or } V_{co} < V \end{cases} \quad (3)$$

Where P_R is the wind turbine's rated power, and V_{ci} and V_{co} are the cut-in speed and cut-out speed, respectively. V_r represents the rated speed of the turbine (Zaki Diab et al., 2019), (Alilou et al., 2025).

3.3.3. Battery modelling

Energy storage is required to maintain a steady, stable power flow to the loads. The lead-acid battery type is proposed for this study. The state of charge of the battery (SOC), charging energy ($E_C(t)$), and discharging energy ($E_D(t)$) can be calculated at any time (t) as such:

1. Charging mode

$$E_C(t) = \left(\frac{P_W(t) - P_{load}(t)}{\eta_{conv}} + P_{PV}(t) \right) \times \Delta t \times \eta_C \quad (4)$$

$$SOC(t) = SOC(t-1)(1 - \sigma) + E_C(t) \quad (5)$$

2. Discharging mode

$$E_D(t) = \left(\frac{P_{load}(t) - P_W(t)}{\eta_{conv}} - P_{PV}(t) \right) \times \Delta t \times \eta_D \quad (6)$$

$$SOC(t) = SOC(t-1)(1 - \sigma) - E_D(t) \quad (7)$$

Where $SOC(t-1)$ is the state of charge of the battery system at time ($t-1$), σ is the self-discharge rate, $P_{load}(t)$ is the load demand for energy at time t , η_{conv} is the converter efficiency, η_C is the charging efficiency of the battery, and η_D is the discharge efficiency of the battery storage system.

3.3.4. Diesel generator modelling

Diesel generator (DG), a non-renewable energy source, is primarily used as a backup in renewable energy microgrids to supplement power shortages from wind and PV systems and meet load demand. The fuel consumption for the DG, $D_f(t)$, can be obtained in hourly intervals by (Zaki Diab et al., 2019):

$$D_f(t) = \alpha_{DG} \times P_{DG}(t) + \beta_{DG} \times P_{DG-R} \quad (8)$$

where $P_{DG}(t)$ is the average power output of the DG per hour, P_{DG-R} is the rated power of the DG, and α_{DG} and β_{DG} are the coefficients of the DG's fuel consumption curve (Zaki Diab et al., 2019).

3.3.5. Energy management strategy

- When the energy produced by the PV and Wind system exceeds the load demand, the batteries will charge from the excess energy. The flow of power can be expressed as:

$$P_{load}(t) \times \Delta t = P_W(t) \times \Delta t + (P_{PV}(t) \times \Delta t - E_C(t)) \times \eta_{conv} \quad (9)$$

- If the energy from the wind and PV is less than the load requirements, then the batteries will discharge to supply the loads as expressed below

$$P_{load}(t) \times \Delta t = P_W(t) \times \Delta t + (P_{PV}(t) \times \Delta t + E_D(t)) \times \eta_{conv} \quad (10)$$

- If the PV, wind, and battery storage cannot meet the load demand, then the diesel generator can cover the generation shortage as follows:

$$P_{DG}(t) \times \Delta t = (P_{load}(t) - P_W(t)) \times \Delta t - (P_{PV}(t) \times \Delta t + E_D(t)) \times \eta_{conv} \quad (11)$$

- The power generated by the PV and wind will be absorbed by a dummy load when the battery reaches maximum charge, as follows:

$$P_{Dummy}(t) \times \Delta t = (P_W(t) - P_{load}(t)) \times \Delta t + (P_{PV}(t) \times \Delta t - E_D(t)) \times \eta_{conv} \quad (12)$$

A proposed operation of the microgrid system is given in Fig. 2.

The control mechanism employed in this study is based on HOMER Pro's built-in optimization and dispatch strategy, which differs from

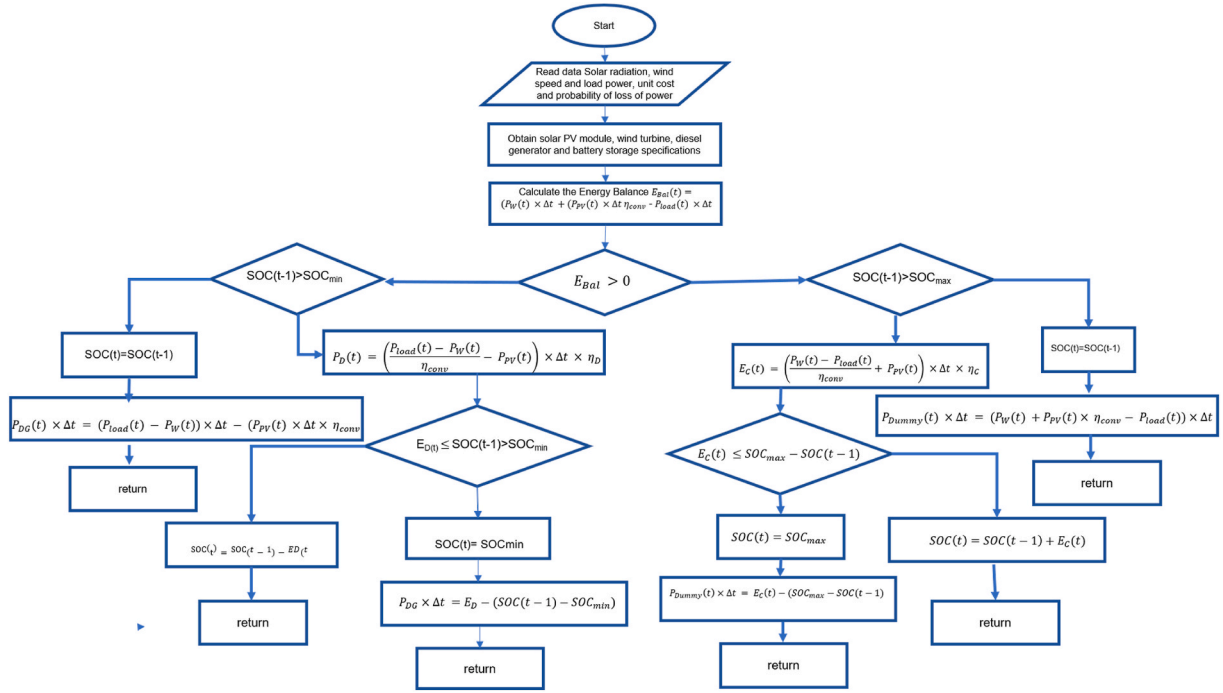


Fig. 2. Proposed microgrid operation (Zaki Diab et al., 2019).

more complex intelligent techniques such as fuzzy logic controllers, Artificial Neural Networks (ANN), Game Theory (GT), Multi-agent system (MAS), and metaheuristic algorithms like genetic algorithms (GA) and Particle Swarm Optimization (PSO) (Seane et al., 2023), (Thirunavukkarasu et al., 2022). The comparison between intelligent techniques (AI and metaheuristics) and HOMER Pro highlights a fundamental difference in their primary focus and scope within Microgrid (MG) Energy Management Systems (EMS). Intelligent techniques generally focus on advanced, dynamic operational optimization and the handling of complexity, whereas HOMER (Hybrid Optimization of Multiple Energy Resources) is widely used as a sophisticated design and techno-economic feasibility tool (Olatomiwa et al., 2016). Table 1. Compares other intelligent optimization and energy management strategy techniques to HOMER Pro.

The novelty of this study lies in its tailored application of HOMER Pro to the Gambian context, integrating site-specific resource data, community load profiles, and local market pricing to simulate and optimize a hybrid microgrid system. This approach improves upon previous solutions by offering a replicable, data-driven control strategy that is both accessible and effective for rural electrification planning. Furthermore, HOMER Pro's integration of technical, economic, and environmental metrics enables a holistic evaluation of system performance, which is often more challenging with standalone intelligent control algorithms (Gautam Rituraj et al., 2024).

3.4. Data collection

Data collection for this study employed primary and secondary data to analyse the feasibility of a renewable energy-based microgrid in Kantong Kunda. Data was collected using Epicollect5 – a mobile and web application for data collection hosted by Imperial College of London. The process involved gathering information on the current energy infrastructure, consumption patterns, future demands, and potential renewable energy resources in Kantong Kunda. Primary data, including historical household energy consumption, hourly energy usage, household size, and appliance power ratings, were collected using structured and semi-structured questionnaires. Secondary data sources, such as Satellite data on energy resources, were obtained from the NASA

Table 1

AI and Metaheuristic intelligent techniques in comparison to HOMER Pro.

Feature	Intelligent Techniques (AI/ Metaheuristics)	HOMER Pro
Primary Focus	Achieving high levels of system reliability and operational efficiency through sophisticated control algorithms (Thirunavukkarasu et al., 2022). Solving complex, non-linear, multi-objective scheduling problems, often emphasizing real-time operational management (Billah et al., 2025), (Allwyn et al., 2023).	Evaluating various design options for hybrid energy systems, primarily through simulation, optimization, and sensitivity analysis (Olatomiwa et al., 2016). Focuses on component sizing for minimum Cost of Energy (COE) production and economic analysis (Olatomiwa et al., 2016).
Optimization Goal	Economic Dispatch (ED), Unit Commitment (UC), Demand Management (DM), and Forecasting (Thirunavukkarasu et al., 2022). Achieving global optima while dealing with complex system characteristics, like nonlinearity and uncertainty (Allwyn et al., 2023).	Determining the optimized size of system components (e.g., PV, battery banks) required to meet the load demand reliably and economically over a long duration (Olatomiwa et al., 2016).
EMS Strategy	Often involves complex mathematical formulations, such as Multi-Agent Systems (MAS), Particle Swarm Optimization (PSO), or Genetic Algorithms (GA), to adapt quickly to changing environments and stochastic inputs (Olatomiwa et al., 2016), (Thirunavukkarasu et al., 2022), (Billah et al., 2025).	Implements EMS by performing hourly energy balance calculations for the entire year (Olatomiwa et al., 2016). The strategy typically treats renewable sources as the primary supply, with the grid or backup sources covering the deficit (Olatomiwa et al., 2016).

Prediction of Worldwide Energy Resource database for resource assessment for the Kantong Kunda community.

3.4.1. Technical design calculations

Table 2 shows the technical calculations for the solar and batteries components of the microgrid.

Table 2

Technical design calculation.

Total power demand (kW)	62.671
Daily energy demand (kWh)	312.927
PV derating factor	0.85
Inverter efficiency	0.95
Battery round-trip efficiency (η_{bat})	0.8
Depth of Discharge (DOD)	0.8
Average Peak sun hours (PSH) (kWh/m ² /day)	5.45
Battery: 12 V, 260 Ah $\Rightarrow$ nominal energy (kWh per battery)	$12 \times \frac{260}{1000} = 3.12$
Overall efficiency (η_{sys})	$0.85 \times 0.95 \times 0.8 = 0.646$
Inverter Capacity (kW): 1.20 x Total power demand (kW)	$1.20 \times 62.671 = 75.205 \approx 80$
Solar PV capacity (P_{PV}) (kW)	$\frac{E_{demand}}{PSH \times \eta_{sys}} = \frac{312.927}{5.451} = 88.88$
Number of 275 W modules	$\frac{88.8745}{0.275} = 323.18 \approx 324$
Installed PV capacity (kW)	$324 \times 0.275 = 89.10$
Battery Bank Capacity (kWh): $E_{bat,nominal} = \frac{E_{demand} \times A}{DOD \times \eta_{bat}}$ considering A = 2 days,	$\frac{312.927 \times 2}{0.80 \times 0.80} = 977.9$
Number of batteries	$\frac{977.896875}{3.12} = 313.43 \approx 314$ batteries

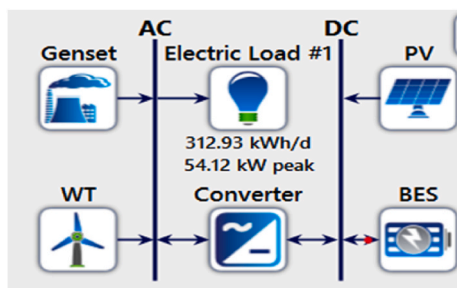
Diesel Generator at full capacity ≈ 80 kW to cover peak load as backup when solar and batteries are unavailable.

3.4.2. Data analysis

HOMER Pro was used to simulate a microgrid system for the Kantong Kunda community, using load data collected during the fieldwork to identify optimal, cost-effective configurations. The simulation results provided a detailed analysis of the potential microgrid configurations. By inputting the load data from the primary and secondary sources, the software simulated and evaluated various combinations of renewable energy sources, energy storage systems, and other components. This approach provided a comprehensive understanding of the technical feasibility of different system configurations to meet the community's energy needs. Through HOMER Pro's optimization feature, the software identified the most cost-effective and reliable equipment combinations for the microgrid system. By considering lifecycle costs, including net present cost, operation and maintenance expenses, and replacement costs, HOMER Pro presented a range of viable system configurations ranked by economic efficiency. This process yielded valuable insights into the economic feasibility of implementing a renewable energy-based microgrid in Kantong Kunda.

3.4.3. Simulation components

In this study, the selection of microgrid components prioritized the available resources in Kantong Kunda and the technological familiarity within the Gambian market. The simulation was configured with the following components: solar PV panels, a wind turbine, a diesel generator, battery storage, and a system converter. Fig. 3 shows the input components for the microgrid design.

**Fig. 3.** Schematic design of input components.

3.4.4. Solar PV

Leveraging Kantong Kunda's abundant solar resources, a PV system was identified as a key component to meet the community's daytime electricity demands. A 275W model type 275YL275D-30b was used for the simulation, with capital and replacement costs of \$95 and an operation and maintenance cost of \$0.10 per year. Table 3 shows the solar module specification.

3.4.5. Wind turbine

A generic 10 kW wind turbine with a hub height of 24m and a 20-year lifespan was simulated. The capital and replacement costs are set at \$50,000, with operation and maintenance costs at \$5/year. Table 4 shows the wind turbine specifications.

3.4.6. Diesel generator

An auto-size diesel generator with a capital and maintenance cost of \$400 was simulated, with a fuel price of \$1/L and an operating and maintenance cost of \$0.02/year. Table 5 shows the diesel generator specification.

3.4.7. Battery storage

A 260Ah, 12V lead-acid battery was used for simulation. The capital cost, replacement costs, and operations and maintenance costs were \$280, \$280, and \$0.28/year, respectively. Table 6 shows the battery storage specifications.

3.4.8. System converter

Since the microgrid integrates both AC and DC components, a bidirectional converter is essential for efficient energy flow. A generic converter with inverter inputs of a 15-year lifetime and efficiency of 95 % and rectifier inputs of 100 % relative capacity and efficiency of 95 % were simulated. The capital and replacement costs were set at \$250, and the operation and maintenance costs at \$2.5/year. Table 7 shows the converter specifications.

3.4.9. Economic inputs

Considering a project lifetime of 25 years and tailoring economic variables to Gambia's context, a discount rate of 17.0 % and an inflation rate of 17.30 % were used for the simulation. These values were based on the Central Bank of The Gambia's December 2023 data. A zero annual capacity shortage is proposed.

3.4.9.1. Economic outputs. HOMER Pro analyzes various system configurations and ranks them based on financial metrics like Net Present Cost (NPC) and COE or LCOE (Cost of Electricity or Levelized Cost of

Table 3

Solar Module specification and cost inputs.

Electrical Parameters at Standard Test Conditions (STC)	
Module type	275YL275D-30b
STC Power Rating	275W
Number of Cells	60
Imp	8.82A
Vmp	31.2V
Isc	9.34A
Voc	38.9V
NOCT	46 °C
Temp. Coefficient of Isc	0.05 %/K
Temp. Coefficient of Power	0.42 %/K
Temp. Coefficient of Voltage	0.124V/K
Maximum System Voltage	1000V
Derating Factor %	85
Cost inputs	
Capacity (kW)	0.275
Capacity (\$)	95
Replacement (\$)	95
O&M (\$/year)	0.1

Table 4
Wind turbine specification and cost inputs.

Wind Turbine Input Specification	
Wind Turbine type	Generic 10 kW
Hub Height (m)	24
Lifetime	20
Cost Inputs (In \$/kW capacity)	
Capital (\$)	50000
Replacement (\$)	50000
O&M (\$/year)	5

Table 5
Diesel Generator specification and cost inputs.

Generator Input Specification	
Generator type	Auto Size
Fuel type	Diesel
Minimum Load Ratio (%)	25
Lifetime	15,000
Cost Inputs (In \$/kW capacity)	
Capital (\$)	400
Replacement (\$)	400
O&M (\$/year)	0.02
Fuel Price (\$/L)	1

Table 6
Battery storage specific and cost inputs.

Battery Storage Specification	
Battery type	Lead-acid
Model	Kinetic Battery model
Nominal voltage (V)	12
Nominal capacity (kWh)	3.1
Maximum Capacity	260
Round-trip efficiency (%)	80
Maximum charge current (A)	43
Minimum state of charge (%)	20
Initial State of Charge (%)	100
Lifetime (years)	20
Throughput (kWh)	3581.60
Cost inputs	
Quantity	1
Capital (\$)	280
Replacement (\$)	280
O&M (\$/year)	0.28

Table 7
Converter specification and cost inputs.

Converter Specification	
Converter type	Generic system
Inverter Input	
Lifetime (years)	15
Efficiency (%)	95
Rectifier Input	
Relative capacity (%)	100
Efficiency (%)	95
Cost inputs	
Capacity (kW)	1
Capital (\$)	250
Replacement (\$)	250
O&M (\$/year)	2.5

Electricity), aiding in selecting the most financially attractive design for the project (Homer Energy, 2016).

4. Results and discussion

4.1. Energy requirement results: energy consumption and demand

4.1.1. Consumption and demand

The combined total daily average consumption of the whole community is 23.0885kWh, with households consuming about 72 %, followed by the school at 15 % and commercial at 10 %. The mosque consumes the least, representing 3 %. Table 8 shows the community's Total Daily Average Energy Consumption. The total energy requirement of the community is 312.927kWh per day. The majority of demand comes from households (82 %), and the mosque accounts for the least, at 2 % of total energy demand. Table 9 shows the community's average total daily energy demand. Fig. 4 shows the Comparison between Daily Average Energy Consumption and Demand. The difference between the household energy consumption and demand is 272.165 kWh. Table 10 illustrates the community's hourly power consumption profile throughout the day and serves as the load input data for the HOMER Pro simulation.

4.2. Simulation results and discussion

HOMER Pro simulated 122,390 solutions; 81,447 were feasible, and 26,840 were infeasible due to the capacity storage constraint. A winning solution was proposed based on optimized architectures with the lowest NPC. The analysis will primarily evaluate the level of renewable energy integration in the proposed solutions. The optimization results will be presented in scenarios that consider renewable energy penetration, along with the overall winning system determined by software analysis. Eleven categories of optimized architectures, shown in Table 11, with different configurations and combinations, were calculated by HOMER Pro. The overall winning system proposed by HOMER Pro for the optimization scenarios was a combination of solar and battery storage systems.

Scenarios:

Scenario 1: 100 % non-renewable using Diesel Generator (Base Case)

Scenario 2: Renewable sources without Diesel Genset

Scenario 3: Renewable and non-renewable configurations

4.2.1. Base case: Diesel Genset only

4.2.1.1. Technical. The base case, featuring a 60 kW generic diesel generator, provides a 100 % non-renewable solution to meet the community's energy needs without capacity shortages. It is designed with a total fuel capacity of 61,157 L, consuming an average of 168 L per day and 6.98 L per hour, resulting in an annual fuel energy input of 601,784 kWh. On average, the generator system is projected to produce 153,654 kWh of electricity annually. Its output ranges from 15 kW to 58 kW, with an average operational output of 17.5 kW. The marginal generation cost

Table 8
Total Daily Average Energy Consumption of the community.

Total Daily Average Energy Consumption of the Community			
No	Description	Energy (kWh)	percentage (≈%)
1	Total Household Energy Consumption	16.739	72
2	Total Commercial Energy Consumption	2.280	10
3	Total Mosque Energy Consumption	0.6195	3
4	Total School Energy Consumption	3.450	15
Total Energy Consumption for the community (kWh)		23.0885	100

Table 9

Average Total Daily Energy Demand of the Kantong Kunda community.

Average Total Daily Energy Demand of the community			
No	Description	Energy (kWh)	percentage (≈%)
1	Total Household Energy Demand	256.505	82
2	Total Commercial Energy Demand	28.2	9
3	Total Mosque Energy Demand	5.382	2
4	Total School Energy Demand	22.84	7
Total Energy Demand for the community (kWh)		312.927	100

stands at \$0.236 per kWh.

4.2.1.2. Economic. The total capital and replacement cost for the generator is \$24,000, calculated at \$400 per kW. Over the project's 25-year lifespan, the generator is expected to operate for approximately 1.71 years, equivalent to around 15,000 cycles. During this period, an estimated 15 replacements will be required, totalling \$347,288.08. Additionally, the overall operation and maintenance cost is estimated at \$271,742.37, alongside a fuel cost of \$1,580,948.37, assuming a fuel price of \$1 per litre. The system has a Net Present Cost (NPC) of \$2,213,744 and an annual operational cost of \$84,707.41, with a LCOE of \$0.7498/kWh.

4.2.1.3. Environmental. Simulation results indicate that using only the diesel generator to meet the community's load demand will result in approximately 160,085 kg of carbon dioxide emissions per year, accounting for 98 % of total emissions. The second-highest emission is carbon monoxide, at 1009 kg per year, followed by nitrogen oxide at 948 kg per year. Sulphur dioxide, unburned hydrocarbons, and particulate matter each account for less than 1 % of overall emissions.

4.2.2. Solar and storage (winning system)

4.2.2.1. Technical. Fig. 5 shows the schematic of the Solar PV and lead-acid battery storage system. The Solar PV and battery architecture requires a converter to supply the loads. This system consists of 275W PV panels, 12V batteries, and a generic 49.9 kW system converter. The solar PV has a rated capacity of 137 kW, a mean output of 24.8 kW, and a daily output of 596kWh, and can produce 217,660kWh of energy yearly. It is expected to operate 4347 h yearly with 191 % PV penetration and a maximum output of 116 kW. Fig. 6 illustrates monthly solar electric production with a battery storage system.

The system requires 544 batteries in 272 parallel strings, each containing two batteries. It is expected to provide 104 h of autonomy and a nominal capacity of 1694 kWh, with 1355 kWh useable. The system has

a lifespan of 19.9 years and will deliver 1,948,390 kWh over its lifetime. With an input energy of 109,068 kWh per year, it can produce 87,358 kWh, resulting in a loss of 21,826 kWh and a storage depletion of 116 kWh. Fig. 7 illustrates the battery's state of charge throughout the year.

The 49.9 kW system converter, which has only an inverter output, has a mean output of 13 kW, a minimum of 0 kW, and a maximum of 49.9 kW. It is expected to operate 8759 h annually, with an input energy of 120,166 kWh/year and an output of 114,158 kWh/year, resulting in losses of 6008 kWh/year. Fig. 8(a) shows the converter's inverter and rectifier outputs. The inverter shows peak power output around 7 a.m.–8 a.m. throughout the year, while the highest average output occurs from 6 p.m. to midnight, with slightly decreased activity between midnight and 6 a.m. in the summer months. This pattern is due to the cyclic nature of solar panels: they operate during the day and store charge in the batteries, serving loads at night when there is no sun. The inverter output of 10 kW during sunny periods helps stabilize the solar output and supply to the loads.

One-day HOMER Pro simulation results (June 2–3, 2023) showing inverter and rectifier power output for the PV–battery system in Fig. 8 (b). The inverter output fluctuates between 5 and 20 kW from 9:00 to 17:00. It remains around 10–15 kW, indicating it is feeding night-time loads from the battery after 20:00 h. The next morning, from 03:00 to 05:00, the output decreases as the battery's state of charge (SoC) declines.

4.2.2.2. Economics. This system configuration has a total NPC of \$273,003.70, an operational cost of \$2257.86, and a LCOE of \$0.09251. The majority of the system's cost, about 71.5 %, is from the battery, with periodic replacements after its lifetime. The total cost of the solar PV is \$48,629.74, about 17.8 % of the cost, with an annual cost of \$1881.18. The total converter cost is about 10.7 % of the system cost, with an annual cost of \$1125.15 shown in Table 12.

The system has a total capital cost of \$214,636.48 and a replacement cost of \$175,864.91. Operating and maintaining this system throughout its lifetime would cost \$9096.86.

Compared with the base system, this configuration has a present worth of \$1,940,741, a 39.2 % return on investment, a 42.9 % IRR, and a simple payback period of 2.32 years. This system generates 34.8 % excess electricity, amounting to 75,783 kWh/year.

4.2.2.3. Environmental. This system design does not produce emissions during electricity generation. Generating power with this system would prevent approximately 160,085 kg of carbon dioxide emissions per year from diesel generators.

This hybrid configuration also offers adequate storage capacity and fewer replacements than the genset. This configuration has cheaper operated and maintenance costs with less pollution. There is potential

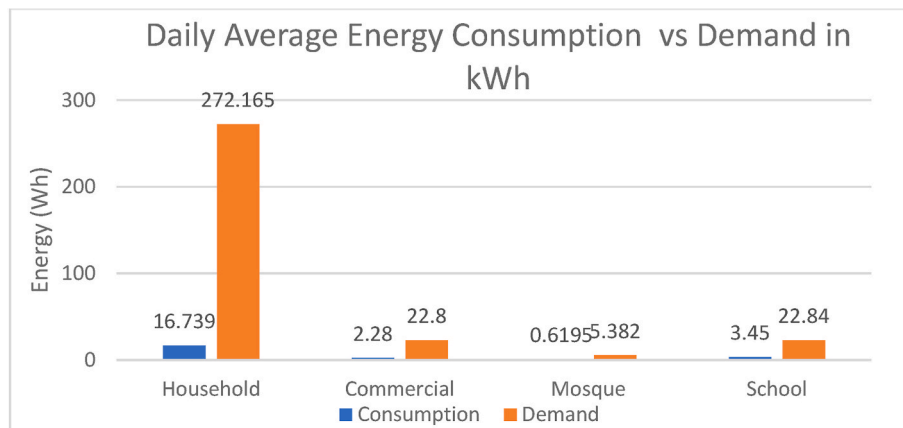


Fig. 4. Comparison between daily average energy consumption and demand.

Table 10

Hourly total load demand of Kantong Kunda community.

Appliance	No.	00:00–01:00	01:00–02:00	02:00–03:00	03:00–04:00	04:00–05:00	05:00–06:00	06:00–07:00	07:00–08:00	08:00–09:00	09:00–10:00	10:00–11:00	11:00–12:00	12:00–13:00	13:00–14:00	14:00–15:00				
Light Bulb (LED)_1	462	1386	1386	1386	1386	1386	1386	1386												
Light Bulb (LED)_2	20	600	600	600	600	600	600	600												
Fan_1	157	11775	11775	11775	11775	11775	11775													
Fan_2	17												1275	1275	1275	1275				
TV	29																			
Radio	29								435						435					
Refrigerator_1	29	4930	4930	4930	4930	4930	4930	4930	4930	4930	4930	4930	4930	4930	4930	4930				
Refrigerator_2	5	750	750	750	750	750	750	750	750	750	750	750	750	750	750	750				
Satellite TV box/ Streaming devices	29																			
Kettle	29								31900											
Mobile Phone Charging_1	165																			
Mobile Phone_2	15										300	300								
Printer	1										50	50								
Speaker	2						120									120				
Radio/ Amplifier	1						30									30				
Computer	2									600	600	600	600	600	600	600				
Total kW		19.441	19.441	19.441	19.441	19.441	19.591	7.666	38.015	6.28	6.63	6.63	7.555	7.555	7.99	7.705				
15:00–16:00	16:00–17:00	17:00–18:00	18:00–19:00	19:00–20:00	20:00–21:00	21:00–22:00	22:00–23:00	23:00–24:00	W _T	hours	Wh									
1275	1275				1386	1386	1386	1386	1386	1386	12	16632								
					600	600	600	600	600	600	12	7200								
			11775	11775	11775	11775	11775	11775	11775	11775	6	70650								
											1275	12	15300							
			4350	4350	4350	4350	4350	4350	4350	4350	4350	6	26100							
4930	4930				435						3	1305								
											4350	6	26100							
											435	3	1305							
											4930	24	118320							
											4930	24	18000							
750	750				750	750	750	750	750	750	24	18000								
					30	30	30	30	30	30	6	5220								
											31900	0.5	15950							
											3300	3	9900							
											300	2	600							
600	600										50	2	100							
											120	3	360							
											30	3	90							
											600	12	7200							
											7.555	7.555	14.11	26.17	24.571	23.971	23.821	23.821	22.741	62.671

Table 11
Optimized system architecture categories.

No						PV (kW)	WT (#)	Genset (kW)	BES (#)	Converter (kW)
1	✓	-	-	✓	✓	137	-	-	544	49.9
2	✓	-	✓	✓	✓	139	-	60	376	42.2
3	✓	✓	-	✓	✓	133	1	-	526	48.4
4	✓	✓	✓	✓	✓	250	1	60	378	37.6
5	-	✓	✓	✓	✓	-	7	60	378	51
6	✓	-	✓	-	✓	354	-	60	-	40.7
7	-	-	✓	✓	✓	-	-	60	118	41
8	✓	✓	✓	-	✓	322	1	60	-	35.1
9	-	✓	✓	-	-	-	5	60	-	-
10	-	-	✓	-	-	-	-	60	-	-
11	-	✓	-	✓	✓	-	64	-	580	82

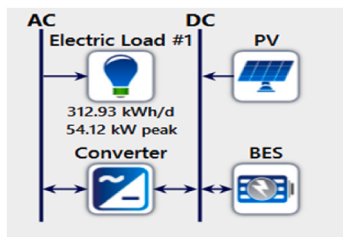


Fig. 5. Solar PV and lead-acid battery storage system schematic.

for agriPV, combining farming and electricity generation on the same piece of land with this proposed system.

4.2.3. All components (hybrid)

The hybrid energy system consisting of 250 kW solar PV, 10 kW wind turbines, a 60-kW diesel generator, 189 strings of lead-acid batteries, and a 37.6 kW converter operating under HOMER Combined dispatch generates 411,145 kWh annually, with solar contributing 96.7 %, wind 3.14 %, and diesel just 0.125 %. Despite high renewable penetration (99.5 %) and excess energy (66.3 %), unmet load is minimal (0.0049 %). The system's Net Present Cost is \$348,800.10, with a levelized cost of \$0.1181/kWh, and a simple payback period of 3.17 years. Financially, it

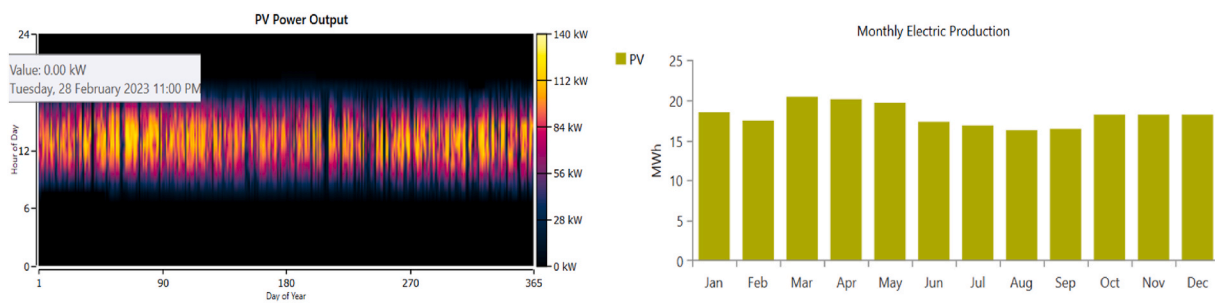


Fig. 6. Monthly electric production from solar with a battery storage system.

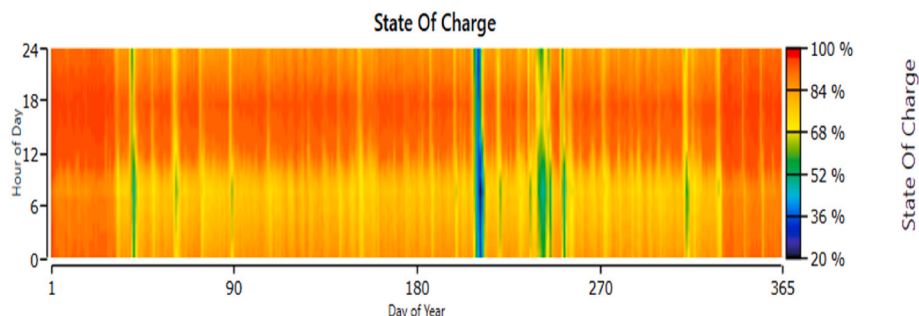


Fig. 7. State of charge of the lead-acid battery throughout the year.

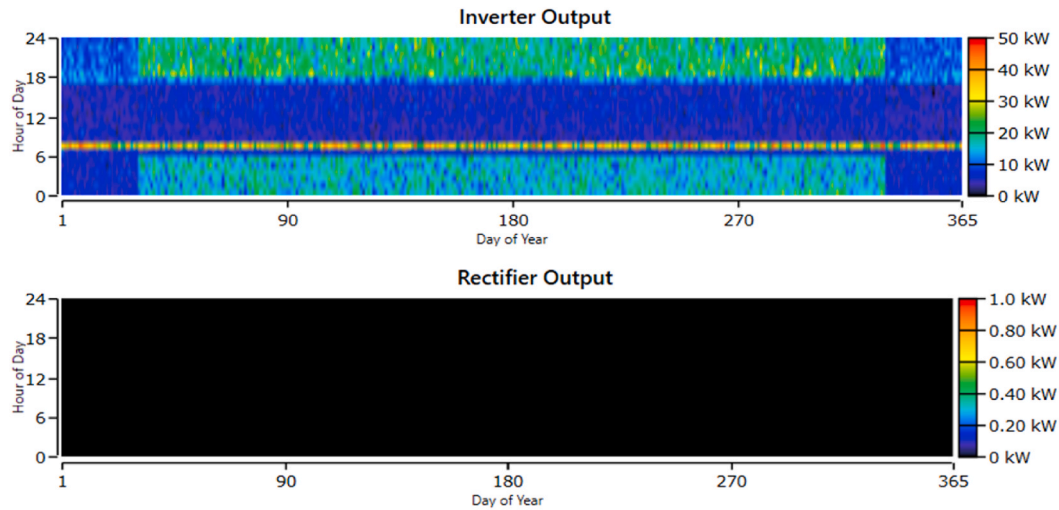


Fig. 8(a). Inverter and rectifier outputs for the solar and battery storage system architecture for one year.

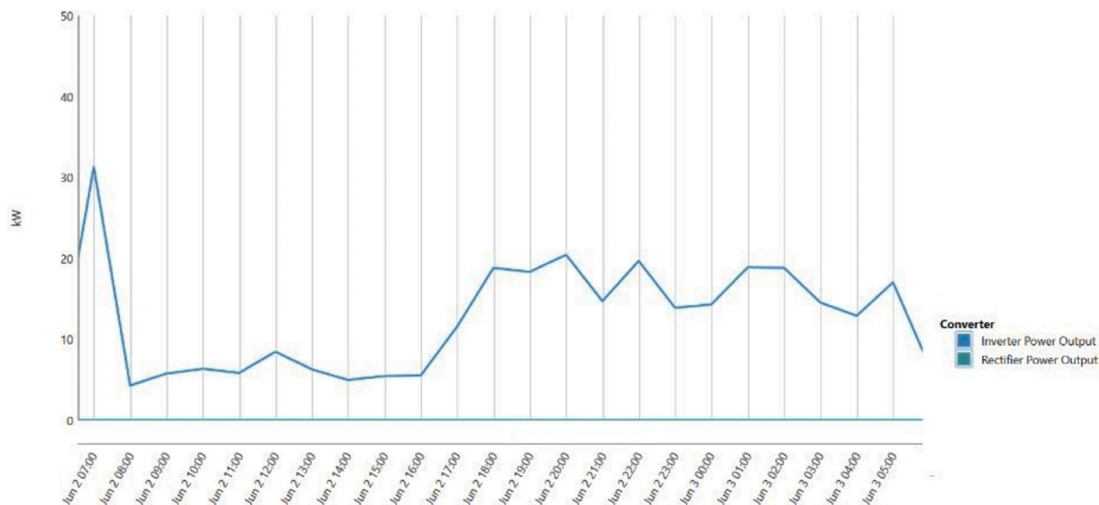


Fig. 8(b). Inverter and rectifier outputs for one day.

Table 12

Cost summary of the PV and battery storage system architecture.

Cost Summary	PV	Lead-acid Battery	System Converter	System
Net Present	\$48,629.74	\$195,288.17	\$29,085.79	\$273,003.70
Annual	\$1881.18	\$7554.47	\$1125.15	\$10,560.79

yields a 28.3 % ROI and 32.2 % IRR, with a present worth of \$1,864,944 compared to the base case. Environmentally, it emits only 400 kg of CO₂ annually, with other pollutants such as CO, NO_x, SO₂, hydrocarbons, and particulates contributing less than 3 kg each, making it a cost-effective, low-emission solution for rural electrification. The proposed solution, incorporating solar energy and battery storage, offers superior economic performance and significantly reduces environmental impact compared to the diesel generator-only scenario and the case combining a diesel generator, PV, wind, and storage.

4.2.4. Sensitivity analysis

A sensitivity analysis was performed on the minimum battery lifetime for the most optimal system. Minimum lifetimes of 5, 8, 10, 15, and

20 years were used as sensitivity variables to assess their effects on overall system cost and configuration. Table 13 presents the simulation results of the proposed system analysis, and Table 14 presents the Sensitivity simulation results for the system architecture and economics.

The system design, configuration, associated costs, and economics change with the minimum battery life. The sensitivity analysis results show that as the minimum battery lifetime increases from 5 to 20 years, the overall system capacity and the renewable fraction increase significantly. The Solar PV capacity of the system increased from 79.8 kW to 139 kW over 5–15 years, then decreased to 133.18 kW for a 20-year minimum lifetime. The number of batteries increased from 374 to 548, and the converter size increased from 22.76 kW to 54.46 kW, enabling greater energy storage capacity. The longer-lasting battery demonstrates more storage capacity and longer autonomy. The shorter battery lifetimes of 5 and 8 years utilize a 60 kW diesel generator to meet the energy demand. However, when the battery life extends from 10 to 20 years, the larger storage capacity enables the system to operate solely on solar energy, eliminating dependence on fossil fuels. Fig. 9 depicts the energy production from the diesel generator and PV.

Extending battery life increases the NPC and LCOE. The analysis shows that the lowest NPC and LCOE occur with a minimum battery lifetime of 5 years, which also has the highest IRR and the shortest

Table 13

Shows the simulation results of the analysis.

Sensitivity/BES Minimum Lifetime (years)	Architecture					System	
	PV (kW)	WT	Gen (kW)	BES (#)	Converter (kW)	Ren Frac (%)	Total Fuel (L/yr)
10	137.04124			544	49.9165328	100	0
15	139.52323			542	50.30679505	100	0
20	133.17771			548	54.45674513	100	0
5	79.81846		60	374	22.76320787	68.29243	10750.52
8	89.520682		60	378	29.14493711	86.11954	5395.376

Table 14

Sensitivity simulation results with system architecture and economics.

Sensitivity/ BES Minimum Lifetime (years)	Cost			Economic	
	Cost/ NPC (\$)	Cost/LCOE (\$/kWh)	Cost/ Operating cost (\$/yr)	IRR (%)	Simple Payback (yr)
10	273003.7	0.09251034	2257.86	10.13329	8.390973
15	273847.5	0.09278578	2274.468	10.10173	8.403481
20	275211.5	0.09326086	2298.882	10.02156	8.427852
5	251474.8	0.08526655	3417.789	14.1541	6.325738
8	254766.5	0.08628482	3298.082	13.48903	6.475832

payback period. However, this results in higher long-term operating costs due to continued dependence on fossil fuels. As the minimum battery lifetime is extended, the IRR decreases and the payback period increases. Thus, while a longer battery life reduces operating expenses and provides a cleaner energy source, it requires a larger initial investment. Fig. 10 demonstrates the trends in NPC and LCOE.

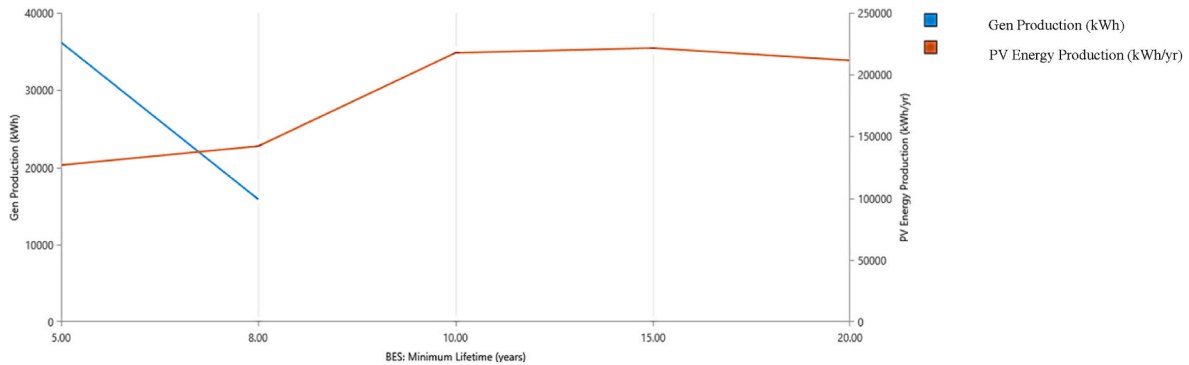
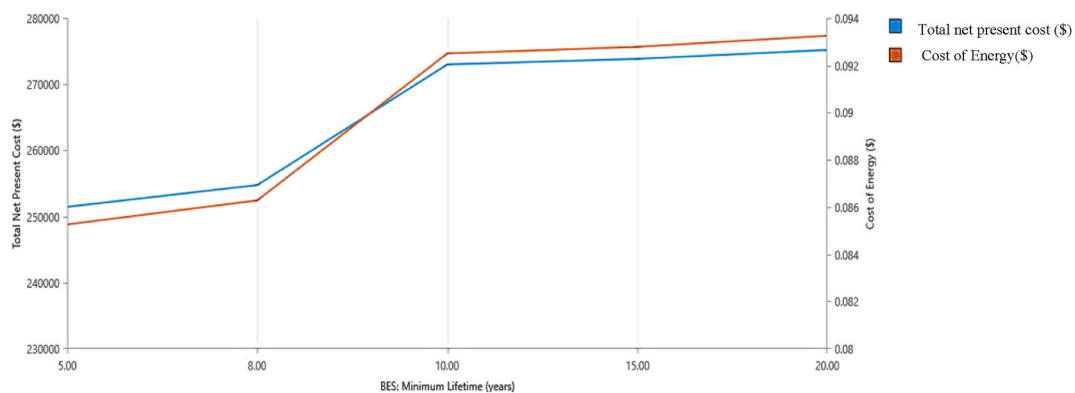
Extending battery life offers environmental benefits by reducing emissions, particularly carbon dioxide. A longer battery life reduces CO₂ emissions and other pollutants by reducing reliance on the diesel

generator, thereby increasing the renewable energy fraction. A 100 % renewable fraction signifies zero emissions, highlighting the positive environmental impact of extending battery life. Fig. 11 shows the renewable energy fraction and carbon dioxide emissions.

Fig. 12 shows the optimization cases for the five-year minimum battery lifetime in the sensitivity case. With LCOE lower than grid electricity (about \$0.23/kWh for domestic consumption) and renewable fractions above 50 %, scenarios two and three offer opportunities for cheaper electricity with increased renewable energy integration.

This study analysed simulation results primarily focused on system architecture, economics, and emissions. The base system chosen for analysis was the 100 % non-renewable diesel generator option. By comparing all other systems with the base system, we gained insights into the effects of the renewable energy fraction on the microgrid's economic and environmental aspects. System designs that are cheaper than the grid tariffs and have a renewable fraction of at least 60 % include:

- Solar + Gen + Battery (Scenario 3)
- Solar + Battery (Scenario 2)
- Solar + Wind + Gen + Battery (Scenario3)
- Solar + Wind + Battery (Scenario2)

**Fig. 9.** Energy production from the diesel generator and PV.**Fig. 10.** Total net present cost and cost of energy.

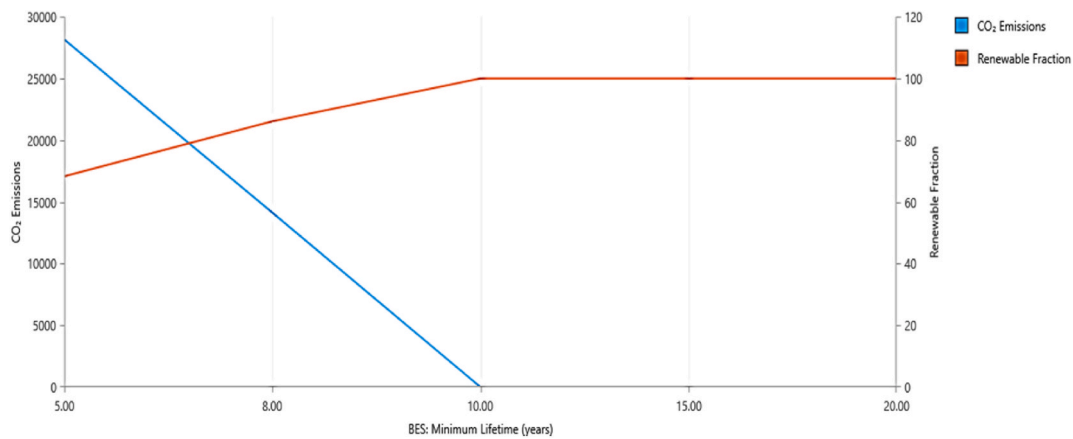


Fig. 11. Renewable fraction and carbon dioxide emission.

Architecture							Cost				System	
     	PV (kW)	WT	Gen (kW)	BES (#)	Converter (kW)	Dispatch	NPC (\$)	LCOE (\$/kWh)	Operating cost (\$/yr)	CAPEX (\$)	Ren Frac (%)	Total Fuel (L/yr)
     	79.8		60.0	374	22.8	CD	\$251,475	\$0.0853	\$3,418	\$163,123	68.3	10,751
     	137			544	49.9	CC	\$273,004	\$0.0925	\$2,258	\$214,636	100	0
     	84.8	1	60.0	376	23.9	CD	\$300,024	\$0.102	\$3,260	\$215,739	78.6	7,242
     	133	1		526	48.4	CC	\$326,665	\$0.111	\$2,661	\$257,875	100	0
     			60.0	112	39.1	CD	\$366,056	\$0.124	\$11,565	\$67,007	0	40,274
     		1	60.0	106	37.9	CD	\$400,525	\$0.136	\$11,044	\$115,003	0	36,171
     	86.4		60.0		16.2	CD	\$423,669	\$0.143	\$14,118	\$50,741	10.7	39,162
     	63.4	1	60.0		14.7	CD	\$476,527	\$0.161	\$14,553	\$100,300	13.3	38,450
     			60.0			CC	\$632,796	\$0.214	\$23,550	\$24,000	0	61,157
     		1	60.0			CD	\$663,466	\$0.225	\$22,803	\$74,000	0	57,685
     		64		580	82.0	CC	\$4.27M	\$1.45	\$34,302	\$3.39M	100	0

Fig. 12. Minimum Lifetime of 5 years optimization configurations.

Microgrids incorporating wind turbines have high costs and make little contribution to the renewable fraction, except when they are the sole energy generator. The configurations involving solar PV, a wind turbine, and a diesel generator, as well as solar PV and a diesel generator, offer competitive electricity costs. Nevertheless, they exhibit renewable energy fractions of 13.3 % and 10.7 %, respectively, indicating a heightened reliance on the diesel generator, which, in turn, will eventually lead to increased emissions.

The most practical microgrid option based on the 5-year battery life (Scenario 3) is also the most optimal case proposed by Homer overall after the sensitivity analysis. It combined solar PV panels (79.8 kW) with a smaller battery bank (374 batteries), a converter (22.8 kW), and a backup diesel generator (60 kW). While this hybrid approach utilized a diesel generator, it still offered significant advantages. It achieved an even lower LCOE of \$0.08527/kWh, a respectable payback period of 6.3 years, and drastically reduced carbon dioxide emissions (less than 50,000 kg annually compared to the baseline's 160,085 kg). Moreover, this system generated excess electricity (16.1 % of total generation) that could be used for future community growth or sold back to the grid when available. Although this system configuration is the most common option, it poses a significant challenge to the 5-year battery life design due to its environmental impact at the end of life and disposal difficulties. Mitigating this challenge is crucial for implementation. Proper disposal methods must be explored. Additionally, exploring alternative designs that balance lifespan, material usage, and recyclability could be a viable solution.

While the solar PV with short-life battery and diesel generator combination stands out as the most practical option, other hybrid designs using wind turbines and storage also showed promise. However,

designs that rely heavily on renewables and have no storage faced potential stability challenges. Microgrids incorporating solar PV and battery storage (or hybrid designs like the one proposed) have the potential to offer significantly cheaper electricity compared to the national grid, particularly for remote communities in The Gambia. Table 15 shows the Comparison of the proposed system and existing systems. Various microgrid system configurations incorporating different levels of renewable energy have been proposed across diverse global regions. As illustrated in Table 15, hybrid generation systems generally yield more favourable cost of energy (COE) values (\$/kWh), whereas Malawi, relying solely on solar PV, exhibits a significantly higher COE. The proposed Solar PV, diesel generator, and battery storage solution with 68.3 % renewable energy contribution performs better than most solutions cited below with respect to COE and would also contribute less CO₂ than solutions with 50 % or less renewable energy contribution, given the same system size.

The manual sizing above used static values for daily energy demand, average peak sun hours, and fixed efficiency factors to estimate system components, often with conservative safety margins. In contrast, HOMER performs an hour-by-hour simulation using the actual load profile, solar resource variability, and component performance under dynamic conditions. This approach allows HOMER to optimize system sizing based on real-time interactions between generation, storage, and load, often resulting in more efficient or cost-effective configurations that may differ from manually derived estimates.

5. Conclusion

This study evaluated the technical, economic, and environmental

Table 15

Studies that compare existing and proposed PV systems.

Reference	Location of Research Study	Configuration of system arrangement	Type of the power generation system	Renewable contribution (%)	NPC(\$)	COE (\$/kWh)
Barber et al. (2019)	Switzerland	wind and solar energy	Distributed generation	100 %	\$5 million and \$20 million	0.05 \$/kWh and 0.2 \$/kWh
Tran et al. (2021)	The island of Dao in Vietnam	solar PV, batteries, and a diesel generator	Distributed generation	40.6 %	\$ 677,663	\$ 0.193/kWh
Kumar et al. (2015)	Chennai, India	wind and solar PV generation	Distributed generation	50 %	\$604,869	\$0.171 per unit
Ali et al. (2023)	Indonesia's Lombok Island	solar PV system, a battery energy storage system	Distributed generation	100 %	\$136,805.50	\$0.2151/kWh
Manoj Kumar et al. (2020)	South Indian community	solar, diesel generator, and battery storage	Distributed generation	80 %	\$440,038.84	\$0.4157/kWh
YUSUPOV (2023)	Karabuk University, Turkey	Utility, PV system, and generator	Distributed generation	14.8 %	\$1.73M	\$0.284/kWh
Eales et al. (2018)	Malawi	Solar PV	Distributed generation	100 %	Small Grid (\$60,707) Medium Grid (\$126,921) Large Grid (\$148,238)	\$2.44/kWh \$1.64/kWh \$1.33/kWh
Kuznetsov et al. (2019)	Abu-Mongar, Egypt	PV, wind, diesel, and battery bank storage	Distributed generation	63.1 %	\$5418863.798	\$ 0.18500434/kWh
Elkadeem et al. (2022)	Egypt at the Sokhna Industrial Zone in Suez	Solar PV and wind	Distributed generation	52 %	\$17,371,980	\$0.0782/kWh
Proposed System	Kantong Kunda, The Gambia	Solar PV, diesel generator, and battery storage	Distributed generation	68.3 %	\$251,475	\$0.0853/kWh

feasibility of implementing a microgrid system for the Kantong Kunda community in The Gambia, integrating solar PV, a diesel generator, and battery storage. The results demonstrate that the proposed system is not only feasible but also offers significant advantages: it delivers a high renewable energy fraction, generates excess electricity (16.1 % of total output), and achieves favourable financial metrics, including a 10.5 % ROI, 14.2 % IRR, and a simple payback period of 6.03 years. Environmentally, it reduces annual carbon dioxide emissions to below 50,000 kg, contributing to a cleaner energy profile. However, the system also presents certain limitations, such as dependency on diesel fuel for backup, which introduces operational costs and residual emissions, and the potential for energy oversupply that may require additional load management strategies. Additionally, the study relies on HOMER Pro's simplified PV modelling, which assumes the presence of a MPPT and a linear relationship between solar radiation and power output. While this assumption improves simulation accuracy for systems equipped with MPPT, it may not fully capture performance variations in systems without MPPT or those that use different MPPT techniques. Overall, the optimized microgrid offers a cost-effective and sustainable pathway to rural electrification, with clear potential to improve energy access and reduce environmental impact. To maximize its benefits, further investment in data collection, load forecasting, and system optimization using machine learning is recommended. These efforts can support national goals for decentralized energy generation and long-term sustainability in The Gambia.

CRediT authorship contribution statement

Maimuna Drammeh: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ravi Samikannu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Abid Yahya:** Writing – original draft, Validation, Methodology, Conceptualization. **Maruliya Begam Kadarmydeen:** Writing – review & editing, Methodology, Conceptualization. **Kavitha R:** Writing – review & editing, Validation, Methodology. **Kanimozhi Gunasekaran:** Writing – review & editing, Validation, Methodology, Conceptualization. **Meeradevi T:** Writing – review & editing, Validation, Methodology. **Gladys Gamariel:** Writing – review & editing, Validation, Methodology.

Funding

The authors did not receive any funding for the research.

Declaration of competing interest

The authors we don't have any conflicts of interest.

Acknowledgments

The authors acknowledge the African Union and the Pan African University of Water and Energy Sciences (PAUWES), Tlemcen, Algeria, for supporting the research.

Data availability

Data will be made available on request.

References

- Ali, M., et al., 2023. A comparison of grid-connected local hospital loads with typical backup systems and renewable energy system based Ad hoc microgrids for enhancing the resilience of the system. *Energies* 16 (4).
- Alilou, M., Younesi, A., Siano, P., 2025. A sustainable approach to hybrid microgrid design : optimal sizing and energy management. *Renew. Energy* 256 (PC), 124078.
- Allwyn, R.G., Al-Hinai, A., Margaret, V., 2023. A comprehensive review on energy management strategy of microgrids. *Energy Rep.* 9, 5565–5591.
- Alshakhs, R., Arefifar, S.A., 2020. Oakland university as A microgrid - feasibility studies of planning and operation. *IEEE Int. Conf. Electro Inf. Technol.* 394–401, 2020-July.
- Barber, S., Boller, S., Nordborg, H., 2019. Feasibility Study for 100 % Renewable Energy Microgrids in Switzerland, pp. 1–16. December.
- Bellagamba, A., 2020. The Gambia. *Africa Yearb* 16 (April), 82–87.
- Billah, M., et al., 2025. Results in engineering decentralized multi-agent control for optimal energy management of neighborhood based hybrid microgrids in real-time networking. *Results Eng.* 27 (February), 106337.
- Blodgett, C., Maret, D., 2015. Nama Design Document for RURAL ELECTRIFICATION WITH RENEWABLE ENERGY IN THE GAMBIA.
- Chandak, S., Rout, P.K., 2021. The implementation framework of a microgrid: a review. *Int. J. Energy Res.* 45 (3), 3523–3547.
- Eales, A., Archer, L., Buckland, H., Frame, D., Galloway, S., 2018. Feasibility study for a solar PV microgrid in Malawi. 2018 53rd Int. Univ. Power. Eng. Conf., pp. 1–6.
- Elkadeem, M.R., et al., 2022. Techno-enviro-socio-economic design and finite set model predictive current control of a grid-connected large-scale hybrid solar/wind energy system: a case study of Sokhna Industrial zone, Egypt. *Energy* 289, 129816, 2024.
- Gautam Rituraj, P.B., Ortiz, Jorge, Ficarella, Niccolo, Chandra Mouli, Gautham Ram, Rodden, Paul, Martell, Christopher, Vallve, Xavier, 2024. Evaluation of Software

- Tools for Standalone Microgrid Design and Optimization 2024 Task 18 Off-Grid and edge-of-grid Photovoltaic Systems. IEA PVPS T.
- Hartono, B.S., Budiyanto, Setiabudy, R., 2013. Review of microgrid technology. 2013 Int. Conf. Qual. Res. Qir 2013 - Conjunction with ICCS 2013 2nd Int. Conf. Civ. Sp., pp. 127–132, 2013.
- Homer Energy, 2016. HOMER Pro Version 3.7 User Manual.
- Hossein Shahinzadeh, G.B.G., Hamid Fathi, S., moazzami, Majid, 2016. Optimal Sizing and Energy Management of a Grid- Connected Microgrid Using HOMER Software, pp. 20–21.
- International Energy Agency, 2022. International Energy Agency (IEA) World Energy Outlook 2022. Int. Energy Agency 524.
- International Trade Administration. *Gambia, the - country commercial guide*. <https://www.trade.gov/country-commercial-guides/gambia-energy>. (Accessed 18 December 2023).
- IRENA, 2018. West African Power Pool: Planning and Prospects for Renewable Energy. Jirdehi, M.A., Tabar, V.S., Ghassemzadeh, S., Tohidi, S., 2020. Different aspects of microgrid management: a comprehensive review. *J. Energy Storage* 30 (April), 101457.
- Kowalczyk, A., Tarnawskiz, J., Włodarczyk, A., 2015. Microgrid energy management system control strategy. *Proc. Conf. Ind. Commer. Use Energy, ICUE 2015-Sept*, 147–154.
- Kumar, A., Abhitharan, K.P., Kalaivani, A., Karthik, T.J., 2015. Feasibility Study of Microgrid Installation in an Educational Institution with Grid Uncertainty, vol. 70, pp. 550–557.
- Kuznetsov, O.N., Sultan, H.M., Aljendy, R.I., Zaki Diab, A.A., 2019. Economic feasibility analysis of PV/wind/diesel/battery isolated microgrid for rural electrification in South Egypt. *Proc. 2019 IEEE Conf. Russ. Young Res. Electr. Electron. Eng. ElConRus* 1001–1006, 2019.
- Lahmeyer International, 2006. Republic of the Gambia. Renewable Energy Study Draft RE Master Plan – Module II.
- Lambert, T., Gilman, P., Lilienthal, P., 2006. Micropower system modeling with homer. *Integr. Altern. Sources Energy* 379–418.
- Liu, B., Qian, X., Li, J., Zhang, Y., 2017. Optimal sizing of distributed generation based on chaotic free-search algorithm in an island microgrid. *Proc. - 2017 Chinese Autom. Congr. CAC 2017-Janua*, 7103–7106, 2017 515770.
- Madduri, P.A., Poon, J., Rosa, J., Podolsky, M., Brewer, E.A., Sanders, S.R., 2016. Scalable DC microgrids for rural electrification in emerging regions. *IEEE J. Emerg. Sel. Top. Power Electron.* 4 (4), 1195–1205.
- Manoj Kumar, N., Chopra, S.S., Chand, A.A., Elavarasan, R.M., Shafiullah, G.M., 2020. Hybrid renewable energy microgrid for a residential community: a techno-economic and environmental perspective in the context of the SDG7. *Sustain. Times* 12 (10), 1–30.
- Marong, L.K., Jirakiattikul, S., anan Techato, K., 2018. The Gambia's future electricity supply system: optimizing power supply for sustainable development. *Energy Strategy Rev.* 20, 179–194.
- MOPE, 2021. Universal Access by 2025 and Transforming the Gambia Electricity Sub-sector no. November.
- Murty, V.V.V.S.N., Kumar, A., 2020. Optimal energy management and techno-economic analysis in microgrid with hybrid renewable energy sources. *J. Mod. Power Syst. Clean Energy* 8 (5), 929–940.
- Nnaji, E.C., Adgidzi, D., Dioha, M.O., Ewim, D.R.E., Huan, Z., 2019. Modelling and management of smart microgrid for rural electrification in Sub-Saharan Africa: the case of Nigeria. *Electr. J.* 32 (10).
- Olatomiwa, L., Mekhilef, S., Ismail, M.S., Moghavvemi, M., 2016. Energy management strategies in hybrid renewable energy systems: a review. *Renew. Sustain. Energy Rev.* 62, 821–835.
- Qureshi, Z. ul A., Kazmi, S.A.A., Mushtaq, S., Anwar, M., 2024. An integrated assessment framework of renewable based microgrid deployment for remote isolated area electrification across different climatic zones and future grid extensions. *Sustain. Cities Soc.* 101, 105069, July 2023.
- Sabzpoosh Saravi, V., Kalantar, M., Anvari-Moghaddam, A., 2024. A cooperative resilience-oriented planning framework for integrated distribution energy systems and multi-carrier energy microgrids considering energy trading. *Sustain. Cities Soc.* 100 (September 2023), 105039.
- Seane, T.B., et al., 2023. Modelling and optimizing microgrid systems with the utilization of real-time residential data: a case study for Palapye, Botswana. *Front. Energy Res.* 11 (March), 1–19.
- Singh, G., Nouhou, S., Sokona, M.Y., 2013. Renewables Readiness Assessment: the Gambia, pp. 1–84.
- Tariq, A.H., Kazmi, S.A.A., Hassan, M., Muhammed Ali, S.A., Anwar, M., 2024. Analysis of fuel cell integration with hybrid microgrid systems for clean energy: a comparative review. *Int. J. Hydrogen Energy* 52, 1005–1034.
- Thirunavukkarasu, G.S., Seyedmehmoudian, M., Jamei, E., Horan, B., Mekhilef, S., Stojcevski, A., 2022. Role of optimization techniques in microgrid energy management systems—A review. *Energy Strategy Rev.* 43 (June 2021).
- Tran, Q.T., Davies, K., Sepasi, S., 2021. Isolation Microgrid Design for Remote Areas with the Integration of Renewable Energy: a Case Study of Con Dao Island in Vietnam. *Cleanroom Technol.* 3 (4), 804–820.
- Veilleux, G., et al., 2020. Techno-economic analysis of microgrid projects for rural electrification: a systematic approach to the redesign of Koh Jik off-grid case study. *Energy Sustain. Dev.* 54, 1–13.
- Xu, D., Numata, M., Mogi, G., 2019. Economic comparison of microgrid systems for rural electrification in Myanmar. *Energy Proc.* 159, 309–314.
- YUSUPOV, Z., 2023. Techno-economic and environmental analysis of microgrid: a case study of Karabuk university. *Sigma J. Eng. Nat. Sci. – Sigma Mühendislik ve Fen Bilim. Derg.* 41 (4), 758–769.
- Zaki Diab, A.A., Sultan, H.M., Mohamed, I.S., Kuznetsov Oleg, N., Do, T.D., 2019. Application of different optimization algorithms for optimal sizing of pv/wind/diesel/battery storage stand-alone hybrid microgrid. *IEEE Access* 7, 119223–119245.
- Zhang, T., Shi, X., Zhang, D., Xiao, J., 2019. Socio-economic development and electricity access in developing economies: a long-run model averaging approach. *Energy Policy* 132, 223–231. October 2018.
- Zia, M.F., Elbouchikhi, E., Benbouzid, M., 2018. Microgrids energy management systems: a critical review on methods, solutions, and prospects. *Appl. Energy* 222 (May), 1033–1055.